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A BANK WANTS A SYSTEM THAT: • STORES CUSTOMER INFORMATION • MANAGES BANK ACCOUNT OPERATIONS • ALLOWS DEPOSIT, WITHDRAWAL, AND BALANCE CHECKING WE WILL USE TWO CLASSES: 1. CUSTOMER 2. BANK ACCOUNT

Main.java	Output
<pre>1- import java.util.Scanner; 2 3 // ----- CUSTOMER CLASS ----- 4 class Customer { 5 String name; 6 int id; 7 8 Customer(String name, int id) { 9 this.name = name; 10 this.id = id; 11 } 12 13 void showCustomerInfo() { 14 System.out.println("Customer Name: " + name); 15 System.out.println("Customer ID: " + id); 16 } 17 } 18 19 // ----- BANK ACCOUNT CLASS ----- 20 class BankAccount { 21 double balance = 0; 22 23 void deposit(double amount) { 24 balance += amount; 25 System.out.println("Deposited: " + amount); 26 } 27 28 void withdraw(double amount) { 29 if (amount <= balance) { 30 balance -= amount; 31 System.out.println("Withdrawn: " + amount); 32 } else { 33 System.out.println("Insufficient Balance");</pre>	<pre>Enter Customer Name: Qazi Enter Customer ID: 72389 ----- CUSTOMER DETAILS ----- Customer Name: Qazi Customer ID: 72389 ----- BANK MENU ----- 1. Deposit 2. Withdraw 3. Check Balance 4. Exit Enter choice: 1 Enter deposit amount: 30 Deposited: 30.0 ----- BANK MENU ----- 1. Deposit 2. Withdraw 3. Check Balance 4. Exit Enter choice: 2 Enter withdraw amount: 34 Insufficient Balance ----- BANK MENU ----- 1. Deposit 2. Withdraw 3. Check Balance 4. Exit Enter choice: </pre>

PRIME NUMBER PROGRAM IN JAVA

Main.java	Run	Output
<pre>2 3 public class Main { 4 5 public static void main(String[] args) { 6 7 Scanner sc = new Scanner(System.in); 8 9 System.out.print("Enter a number: "); 10 int n = sc.nextInt(); 11 12 int count = 0; 13 14 if (n <= 1) { 15 System.out.println(n + " is not a Prime Number"); 16 } else { 17 18 for (int i = 1; i <= n; i++) { 19 if (n % i == 0) { 20 count++; 21 } 22 } 23 24 if (count == 2) { 25 System.out.println(n + " is a Prime Number"); 26 } else { 27 System.out.println(n + " is not a Prime Number"); 28 } 29 } 30 31 sc.close(); 32 }</pre>		Enter a number: 13 13 is a Prime Number === Code Execution Successful

FIND LENGTH OF ARRAY

Main.java	   Share	Run	Output
<pre>1- public class Main { 2 3- public static void main(String[] args) { 4 5 int arr[] = {10, 20, 30, 40, 50}; 6 7 System.out.println("Array Elements:"); 8 9- for (int i = 0; i < arr.length; i++) { 10 System.out.println(arr[i]); 11 } 12 13 System.out.println("\nLength of Array: " + arr.length); 14 } 15 }</pre>			Array Elements: 10 20 30 40 50 Length of Array: 5 === Code Execution Successful ===

FIND THE NUMBER IS PALINDROME OR NOT

Main.java	Run	Output
<pre>1 import java.util.Scanner; 2 3 public class Main { 4 5 public static void main(String[] args) { 6 7 Scanner sc = new Scanner(System.in); 8 9 System.out.print("Enter a number: "); 10 int num = sc.nextInt(); 11 12 int original = num; 13 int reversed = 0; 14 15 while (num != 0) { 16 int digit = num % 10; 17 reversed = reversed * 10 + digit; 18 num = num / 10; 19 } 20 21 if (original == reversed) { 22 System.out.println(original + " is a Palindrome Number"); 23 } else { 24 System.out.println(original + " is NOT a Palindrome Number"); 25 } 26 27 sc.close(); 28 } 29 }</pre>		<pre>Enter a number: 12 12 is NOT a Palindrome Number === Code Execution Successful ===</pre>

FIND FACTORIAL OF A NUMBER THROUGH FUNCTION

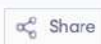
Main.java	   Share	Run	Output
<pre>1 import java.util.Scanner; 2 3 class Factorial { 4 5 int fact(int n) { 6 int result = 1; 7 8 for (int i = 1; i <= n; i++) { 9 result = result * i; 10 } 11 12 return result; 13 } 14 } 15 16 public class Main { 17 18 public static void main(String[] args) { 19 20 Scanner sc = new Scanner(System.in); 21 22 System.out.print("Enter a number: "); 23 int num = sc.nextInt(); 24 25 Factorial f = new Factorial(); 26 27 int result = f.fact(num); 28 29 System.out.println("Factorial of " + num + " is: " + result); 30 31 sc.close();</pre>			<pre>Enter a number: 12 Factorial of 12 is: 479001600 === Code Execution Successful ===</pre>

SWAPPING NUMBERS USING TWO VARIABLES

Main.java	   Share	Run	Output
<pre>1- import java.util.Scanner; 2 3- public class Main { 4 5- public static void main(String[] args) { 6 7 Scanner sc = new Scanner(System.in); 8 9 System.out.print("Enter first number (a): "); 10 int a = sc.nextInt(); 11 12 System.out.print("Enter second number (b): "); 13 int b = sc.nextInt(); 14 15 System.out.println("\nBefore Swapping:"); 16 System.out.println("a = " + a); 17 System.out.println("b = " + b); 18 19 // Swapping logic (without third variable) 20 a = a + b; 21 b = a - b; 22 a = a - b; 23 24 System.out.println("\nAfter Swapping:"); 25 System.out.println("a = " + a); 26 System.out.println("b = " + b); 27 28 sc.close(); 29 } 30 } 31</pre>			<pre>Enter first number (a): 12 Enter second number (b): 23 Before Swapping: a = 12 b = 23 After Swapping: a = 23 b = 12 === Code Execution Successful ===</pre>

RIGHT ANGLED TRIANGLE PATTERN USING NESTED LOOP

Main.java



Run

Output

```
1 import java.util.Scanner;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter number of rows: ");
10        int rows = sc.nextInt();
11
12        for (int i = 1; i <= rows; i++) {
13
14            for (int j = 1; j <= i; j++) {
15                System.out.print("* ");
16            }
17
18            System.out.println();
19        }
20
21        sc.close();
22    }
23 }
```

```
Enter number of rows: 5
*
* *
* * *
* * * *
* * * * *
```

=== Code Execution Successful ===